

8-Bit Signed Binary Number Representation Using Two's Complement

DESIGN AND SIMULATION IN LOGISIM



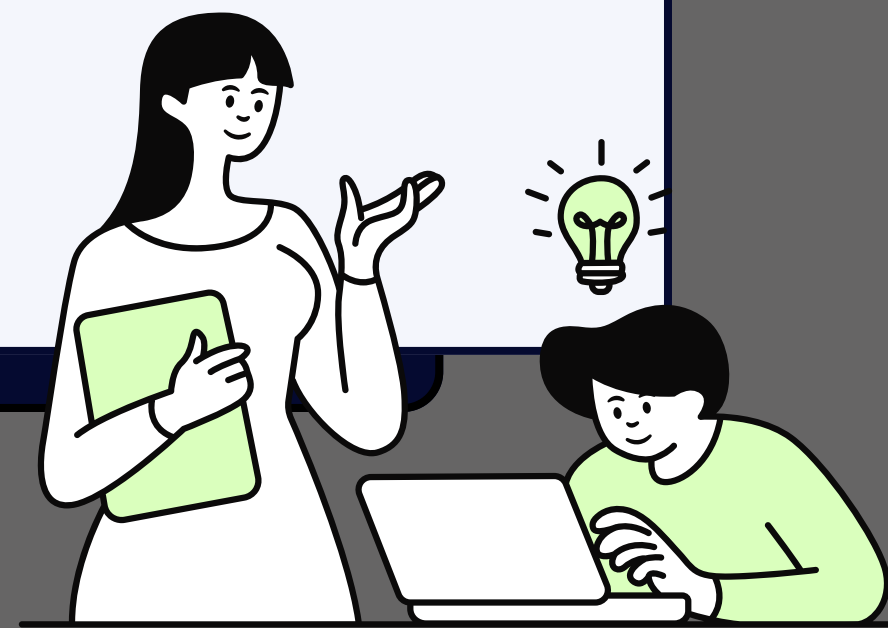
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Problem Statement

Design an 8-bit binary circuit in Logisim

Represent positive and negative numbers.

Perform addition and subtraction.

Use Two's Complement for subtraction.

Design / Methodology

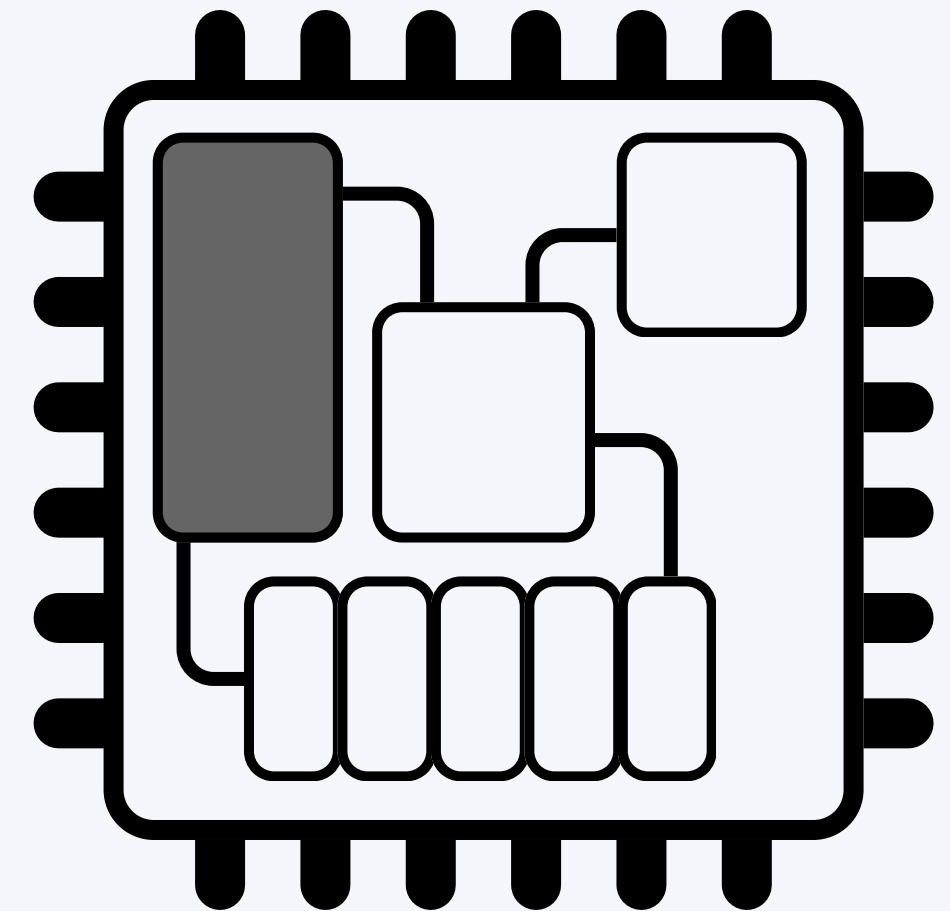
Addition

Two 8-bit numbers are given as input.
Full Adders are used for addition.

Subtraction

Invert the second number using NOT gates.
Add 1 to get Two's Complement.
Add it to the first number.

$$A - B = A + 2's \text{ Complement of } B$$



Range of 8- bit Signed Numbers

Minimum = -128

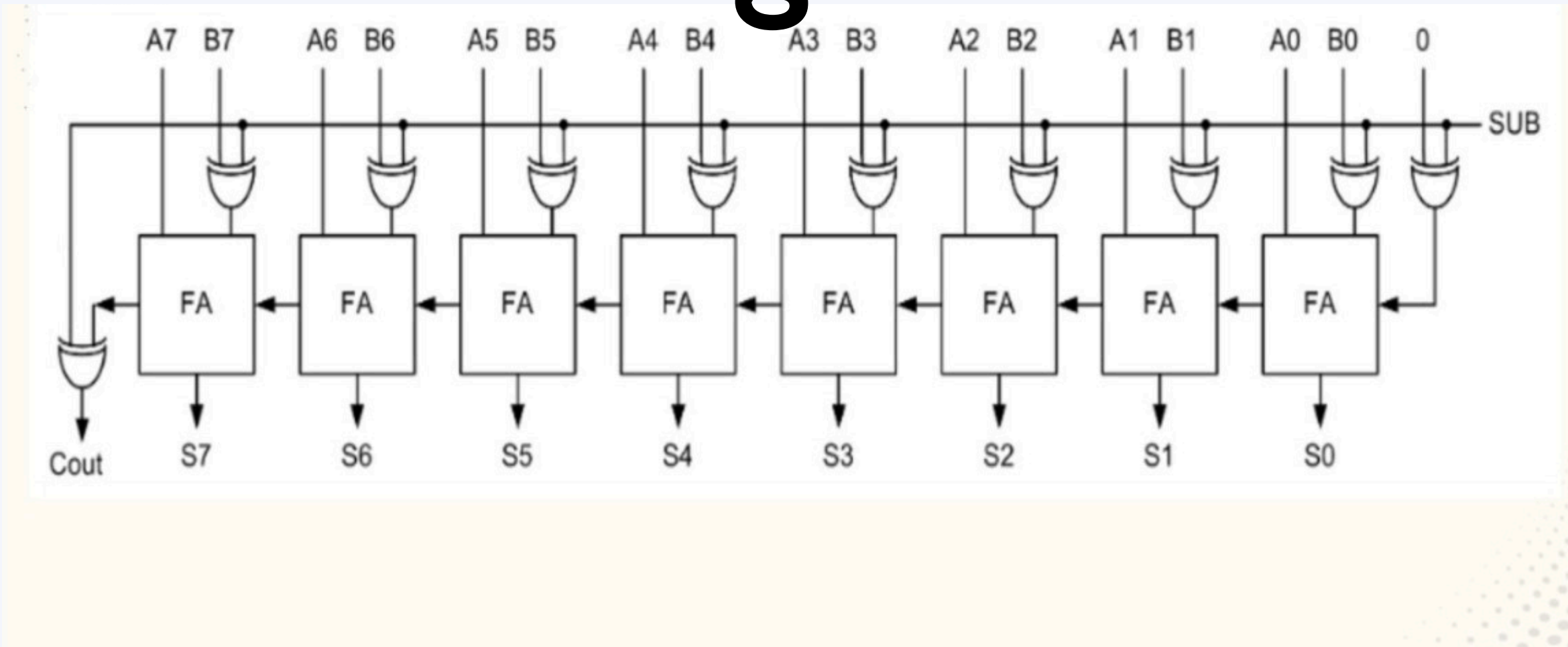
Maximum = +127

Sign value

0 → positive

1 → negative

Demo in Logisim



A =

0	0	0	0	1	0	0	1
---	---	---	---	---	---	---	---

B =

0	0	0	0	0	1	0	1
---	---	---	---	---	---	---	---

Result =

0	0	0		0	0	0	1	1
---	---	---	--	---	---	---	---	---

$$8 - 5 = 3$$

Github repository

<https://github.com/neerudimadhupriya-cmyk>